

ITEC614 - Cyber Security Concepts

Assessment 3: Risk Analysis Report

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Appendix I: Feed Forward

Questions	Feedback recommendation
1. How have you acted on the feedback from the previous assignment to improve your work in this assignment?	Based on my past criticism, I made beyond any doubt that I backed my talk, position, thoughts, and ideas with peer-reviewed journal references in this task.
2. What is your expectation around the type of feedback that enhances your learning?	I need to know where I made a botch and how I can rectify them and not make the same botch once more, i.e., I need criticism that will help me to move forward with my learning and execution within another task.
Did you have any difficulty understanding or acting on previous feedback? Please be as specific as possible so that you can gain further feedback/clarify anything you do not understand in the feedback.	The input given in my past task was exceptionally bland; I did not know how to improve my work. So, I would just like the instructor to clarify more on related subject viewpoints of the input, or I would like an opportunity to have an exchange to get the input.

Appendix II: Assignment Checklist

My submitted assignment report is within the specified word limit.	✓
I have included references using the specified referencing style.	✓
I have correctly cited all my sources and references.	✓
I have formatted my report as per the specifications.	✓
I have checked my Turnitin report to ensure the similarity report is within the acceptable level (below 20% similarity)	✓
I have included specified process output in my submission (e.g., reflection report, e-portfolio submission, video summary, screen dump of my search and retrieval of journal articles, etc.)	✓
I have actioned feedback advice provided to me from previous assignment feedback (if applicable)	✓
I have completed proofreading and checked for spelling and grammar.	✓
I have submitted my work before the due date/time.	✓
I have submitted feed feed-forward template along with my assignment submission.	✓

Risk Analysis and Security Recommendations for a Hotel Management System Using STRIDE and DREAD Models

A.Tamang

Abstract - In the modern hospitality industry, Hotel Management Systems (HMS) serve as the backbone of daily operations, handling tasks ranging from customer bookings and billing to inventory and employee management. This report presents a comprehensive risk analysis of a Hotel Management System using two industry-standard models: STRIDE for identifying and classifying threats, and DREAD for evaluating the severity of those threats. Through this structured approach, five key security threats are identified, each mapped to relevant system vulnerabilities and risk levels. The analysis reveals significant concerns regarding data confidentiality, system availability, and authentication weaknesses that could impact both hotel operations and customer trust. The report concludes by recommending specific mitigation strategies, such as role-based access control, encryption protocols, and intrusion detection systems, to fortify the HMS against future security breaches. This work contributes to the broader conversation on strengthening information systems in the hospitality sector, ensuring both compliance and customer satisfaction through robust cybersecurity planning.

Index Terms - hotel management system, cybersecurity, risk analysis, phishing attacks, integrity

Introduction

As a cybersecurity student and future professional, I've come to understand the vital importance of secure information systems in industries such as hospitality, where customer trust is paramount. For this report, I chose to analyze a Hotel Management System (HMS) because I've personally interacted with it as a customer, and I am aware of the sensitive data it handles, from full names and IDs to credit card details and booking preferences. These systems also often connect to third-party services like payment gateways and online travel agencies, which makes them even more vulnerable if not properly secured.

Hotel Management Systems are designed to streamline operations such as room bookings, check-ins and check-outs, billing, housekeeping coordination, and customer feedback. A typical HMS involves several modules, each interacting with central databases and often accessed by staff from different departments. For instance, front desk staff access booking and customer data, while finance staff manage billing records. This interconnectedness, while efficient, also increases the risk of internal and external security threats.

The importance of securing such a system cannot be overstated. A single breach could not only compromise the personal and

financial data of thousands of guests but also damage the hotel's reputation and result in significant legal and financial consequences. In recent years, cyberattacks on hotel chains like Marriott and MGM Resorts have shown how devastating such incidents can be on a global scale [1].

This report aims to provide a detailed risk analysis of a Hotel Management System using two established security models: STRIDE, which helps classify threats based on security properties, and DREAD, which helps rate the severity of those threats. By using these models, I aim to identify the most critical risks, understand their potential impact, and recommend appropriate security measures that a real-world hotel could implement to protect both its data and its customers.

Analyzing Impacts of Security Threats on Hotel Management Systems

Security threats to Hotel Management Systems (HMS) go far beyond technical glitches; they have real-world consequences that can ripple across borders, industries, and societies. In my analysis, I've found that the impacts of these threats can be grouped into three major areas: operational disruption, data breaches, and reputational damage.

Firstly, from an operational standpoint, any cyberattack that disrupts the HMS, such as a Denial-of-Service (DoS) attack, can bring hotel operations to a halt. For instance, if front desk staff can't access the booking module due to a ransomware attack, it creates

chaos during guest check-ins and outs. In international hotel chains, such disruptions could lead to simultaneous operational failures in multiple countries, especially if systems are centrally managed. This was witnessed in the MGM Resorts cyberattack in 2023, where systems across the U.S. were knocked offline, causing check-in delays, blocked access to rooms, and a complete failure of the loyalty program system. [2].

Secondly, security threats can lead to large-scale data breaches that expose customers' personally identifiable information (PII). Hotel systems store not just names and addresses but also passport numbers, payment information, and stay histories, making them lucrative targets for cybercriminals. When Marriott International was breached in 2018, data of over 500 million guests was compromised, including encrypted credit card information and passport details. [2]. This breach had consequences that went beyond inconvenience; it raised national security concerns in some countries and triggered regulatory investigations under data protection laws like GDPR.

Lastly, the reputation damage from a security threat can be long-lasting. In today's connected world, news of a breach spreads quickly. Customers may avoid staying at a hotel chain if they feel their privacy isn't respected or protected. Negative publicity, lawsuits, and regulatory fines can cripple a brand's trustworthiness and financial standing, particularly in an industry where loyalty and trust are key.

In a global context, these threats also highlight the lack of uniform cybersecurity

regulations across countries. A breach that originates in one country can impact data stored in another. This cross-border nature of cyber threats underlines the importance of international cooperation in cybersecurity frameworks, especially for globally operating hotel chains.

From my perspective, working on this assessment made me appreciate how interconnected and vulnerable information systems are, and how a lack of security planning in one area can create a domino effect of consequences.

Identifying Five Common Security Threats in Hotel Management Systems

While exploring the inner workings of Hotel Management Systems (HMS), I began to realize how many different attack surfaces exist within even a single hotel's digital infrastructure. From what I've researched and understood, five common security threats pose significant risks to HMS platforms, especially those that deal with sensitive customer and financial data. These threats affect key aspects of cybersecurity, including confidentiality, integrity, availability, authentication, and non-repudiation.

1. Unauthorized Access (Threat to Confidentiality & Authentication)

One of the biggest threats I came across is unauthorized access, where attackers or even insiders gain access to modules they shouldn't be allowed to see. For example, a

front desk employee might try to access the financial records module, or worse, an attacker might exploit weak password policies to gain access from outside the network. This type of threat often results from poor access controls or a lack of multi-factor authentication. [3]. Once inside, attackers can view or steal sensitive guest information, violating confidentiality and authentication protocols.

2. SQL Injection Attacks (Threat to Data Integrity)

SQL injections remain one of the oldest yet most effective threats to web-based systems like HMS. I've learned that if an HMS doesn't validate user input properly, attackers can insert malicious SQL code into forms like the booking portal or login screens. This can allow them to modify or delete entire databases, compromising data integrity and even bringing down the entire reservation system. [3]. It's a simple yet powerful method for data corruption and unauthorized manipulation.

3. Denial of Service (DoS) Attacks (Threat to Availability)

During my research, I found that DoS or Distributed DoS (DDoS) attacks are common in hospitality, especially during peak seasons. These attacks flood the HMS servers with fake traffic, making the actual system slow or entirely unavailable for genuine users. In hotels, this could lead to failure in booking systems, digital check-ins, or even locking guests out of smart rooms. These attacks

directly impact the system's availability, a critical aspect of cybersecurity [2].

4. Phishing and Social Engineering (Threat to Non-Repudiation & Authentication)

Human error is often the weakest link. Many hotel staff members, especially seasonal or part-time employees, may fall for phishing emails that appear to come from IT support or management. These emails often contain fake password reset links or malicious attachments. Once clicked, attackers can hijack credentials and impersonate staff, making it hard to trace the source of the actions [4]. This severely undermines non-repudiation and authentication controls.

5. Malware Infections (Threat to Confidentiality & Availability)

Lastly, malware, whether introduced through a USB device or an infected website, is a huge threat. For example, ransomware can encrypt HMS data, effectively locking the hotel out of its system unless a ransom is paid. This not only compromises availability but can also leak confidential customer data during the breach. The hospitality sector has seen multiple ransomware attacks recently, which shows that this threat is far from theoretical. [3].

Analyze Security Requirements Using the STRIDE Model

When I first started analyzing the Hotel Management System through the lens of the

STRIDE model, I realized how effective this framework is in breaking down complex threats into understandable categories. STRIDE, which stands for Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege, allowed me to think more critically about the kinds of security requirements needed to protect an HMS and the real risks behind them.

Below, I've mapped out each STRIDE threat with corresponding security requirements, using what I've learned both in this unit and from academic sources.

1. Spoofing - Implement Strong Authentication Mechanisms

Spoofing occurs when an attacker pretends to be another user, often by stealing login credentials. In the context of an HMS, a cybercriminal could spoof an admin user and gain control over all booking and payment modules. To prevent this, the system must implement strong user authentication mechanisms, such as two-factor authentication (2FA) or biometric logins for staff accessing sensitive modules. [5]. Role-based access controls (RBAC) should also be enforced to ensure users can only access modules relevant to their job roles.

2. Tampering - Use Data Integrity Verification Techniques

Tampering refers to unauthorized modifications of data. A threat actor could alter room rates, change booking records, or manipulate financial logs. To protect against

tampering, the system should use data integrity checks, such as cryptographic hashes and digital signatures, especially during data transmission and storage [5]. Secure API calls with validation routines can also help ensure only authorized changes are made to the database.

3. Repudiation - Enable Logging and Auditing

Repudiation is when a user denies performing an action, such as cancelling a booking or processing a refund. In an HMS, this could cause serious disputes between staff and customers. To mitigate this, comprehensive activity logs must be maintained. These logs should include user IDs, timestamps, and descriptions of actions, and be made tamper-proof. This way, actions can be traced and audited for accountability. [4].

4. Information Disclosure - Encrypt Sensitive Data

Information disclosure involves the exposure of sensitive information to unauthorized users. In hotel systems, this could mean leaked credit card numbers, passport data, or personal preferences. To protect against this, both in-transit and at-rest data should be encrypted using standards like AES-256 or TLS 1.3 [2]. Access to personal data must also be logged and limited to those with valid business reasons.

5. Denial of Service (DoS) - Implement Redundancy and Rate Limiting

DoS attacks aim to make the system unavailable to legitimate users by overwhelming it with traffic or malicious requests. In an HMS, this could mean shutting down the online booking platform during peak travel seasons. To defend against such attacks, hotels should use rate-limiting controls, web application firewalls (WAFs), and scalable cloud infrastructure that can handle spikes in traffic. [5].

6. Elevation of Privilege - Restrict Admin Access and Monitor Anomalies

This occurs when an attacker gains higher access privileges than they're authorized for, like a regular employee somehow getting admin rights. HMS platforms must enforce least-privilege principles, and any privilege escalation requests should be tightly monitored and verified. Tools like Security Information and Event Management (SIEM) systems can also detect abnormal behavior in real time. [1].

Using the STRIDE model helped me identify the core weaknesses of the HMS and design corresponding controls to address them. It made me realize that cybersecurity isn't just about defending against outside threats; it's also about putting the right structures in place to reduce human error and insider risks.

Risk Rating Using DREAD Model

Working through the DREAD model helped me view cybersecurity risks not just as abstract possibilities, but as measurable threats that a real hotel might face. DREAD stands for Damage Potential, Reproducibility, Exploitability, Affected Users, and Discoverability, and each threat can be scored between 1 and 10 in each category. By quantifying these values, I was able to prioritize which threats need the most urgent attention.

Below is how I applied the DREAD model to the five security threats identified earlier for the Hotel Management System (HMS). The total risk score is the average of the five DREAD factors.

1. Unauthorized Access (Spoofing Attack)

This threat occurs when an attacker impersonates a legitimate user, often by stealing login credentials. If an admin account is compromised, the attacker could potentially access all customer data, booking records, and billing information.

- **Damage Potential:** 9
- **Reproducibility:** 7
- **Exploitability:** 8
- **Affected Users:** 9
- **Discoverability:** 6

Average DREAD Score: 7.8

Given its high impact and the large number of affected users, this is one of the top-priority risks for any HMS. [1].

2. SQL Injection

A well-known web application vulnerability, SQL injection, can allow attackers to manipulate the hotel's backend database by injecting malicious commands through user inputs, like the booking or login form.

- **Damage Potential:** 8
- **Reproducibility:** 8
- **Exploitability:** 7
- **Affected Users:** 8
- **Discoverability:** 7

Average DREAD Score: 7.6

Although SQL injection is an old trick, it's still highly effective against poorly coded applications [3].

3. Denial of Service (DoS Attack)

DoS attacks can bring down the HMS, especially the public booking and reservation systems, causing massive disruption during peak periods.

- **Damage Potential:** 7
- **Reproducibility:** 6
- **Exploitability:** 6
- **Affected Users:** 9
- **Discoverability:** 8

Average DREAD Score: 7.2

While it may not always involve data theft, the operational and reputational impact of DoS attacks in the hospitality industry is significant [2].

4. Phishing and Social Engineering

This is where attackers trick hotel staff, usually through fake emails, into giving up passwords or clicking on malicious links. It's surprisingly effective because humans are often the weakest link.

- **Damage Potential:** 6
- **Reproducibility:** 9
- **Exploitability:** 8
- **Affected Users:** 6
- **Discoverability:** 9

Average DREAD Score: 7.6

This threat is very common, and even a single compromised account can lead to larger breaches [4].

5. Ransomware / Malware Infection

Ransomware attacks encrypt the hotel's data, locking out the staff until a ransom is paid. These attacks are increasingly targeting the hospitality sector.

- **Damage Potential:** 10
- **Reproducibility:** 6
- **Exploitability:** 6
- **Affected Users:** 8
- **Discoverability:** 5

Average DREAD Score: 7.0

This threat may have lower reproducibility, but carries devastating damage when successful [5].

Overall Risk Severity of the System

To assess the overall risk to the HMS, I calculated the average DREAD score across all five threats, which comes out to:

$$(7.8+7.6+7.2+7.6+7.0) / 5 = 7.44$$

This shows a high overall risk, confirming that the Hotel Management System is a critical asset that needs robust security controls at every level, from staff training to system hardening and real-time threat detection.

Recommendations

After carefully analyzing the threats facing a Hotel Management System and evaluating their severity through STRIDE and DREAD models, I've come to see that effective cybersecurity is not just about identifying risks about responding to them with practical and sustainable solutions. If I were advising a real hotel, these are the security measures I would strongly recommend based on current best practices and recent research in the field.

1. Implement Role-Based Access Control (RBAC)

One of the first things a hotel should do is implement role-based access control. From my research, I found that many security incidents begin with employees accessing parts of the system they don't need to use.

With RBAC, each staff member only has access to the data and functions required for their specific job [1]. For example, front desk staff should not have access to the billing backend or administrative functions.

2. Adopt Strong Multi-Factor Authentication (MFA)

Passwords alone are no longer enough to secure hotel systems. Adding a second layer, like a one-time code sent to a mobile device or fingerprint scan, can drastically reduce unauthorized access attempts. Implementing MFA, especially for administrative roles, is a proven way to mitigate spoofing attacks. [2].

3. Use Encryption for Data at Rest and in Transit

Given how sensitive data a hotel handles, guest names, credit card details, and passport numbers, encryption should be mandatory. TLS encryption protects data in transit, while AES-256 encryption keeps stored data safe from attackers who manage to gain access to servers [5]. This would help prevent large-scale data leaks even in the event of a breach.

4. Deploy Web Application Firewalls (WAF) and Input Validation

To defend against SQL injections and similar web-based attacks, input validation and WAFs are essential. Web application firewalls filter malicious requests before they reach the system, while proper input sanitization ensures that the system doesn't process harmful commands embedded in

form fields [3]. These defenses are especially important for public-facing modules like online booking forms.

5. Train Staff Against Phishing and Social Engineering

It became clear to me during this project that technical solutions are only half the battle. Employees need to be trained regularly on how to recognize phishing attempts, handle suspicious emails, and follow secure procedures. Simple awareness training can prevent incidents that would otherwise result in credential theft or malware infections. [4].

6. Set Up Regular Backups and Ransomware Recovery Plans

Ransomware is one of the most damaging attacks hotels can suffer from. One way to reduce its impact is to perform regular, automated backups of the HMS database and configuration files. These backups should be stored in secure, offline environments so they can't be encrypted during an attack. Having a recovery plan also helps minimize downtime. [5].

7. Monitor System Activity Using SIEM Tools

Finally, implementing Security Information and Event Management (SIEM) tools can help detect abnormal patterns, such as failed login attempts or strange access times, and trigger alerts in real time. These tools help monitor internal and external threats, making

it easier to respond before serious damage is done [2].

Conclusion

Working on this risk analysis of a Hotel Management System has deepened my understanding of how complex and layered cybersecurity is. At first, I thought of these threats as mainly technical issues, something for IT teams to fix with firewalls and antivirus software. But using structured frameworks like STRIDE and DREAD, I've learned that addressing these threats requires a combination of technical controls, human awareness, and strategic planning.

This report identified five common threats: unauthorized access, SQL injection, denial of service, phishing, and ransomware, and analyzed each one through the STRIDE model to understand what they target: from data confidentiality and integrity to system availability and authentication. Then, using the DREAD model, I rated the severity of these threats and realized that nearly all of them posed a high risk to the HMS, with an average score of 7.44. This made it clear that without strong defenses, hotel systems are highly vulnerable.

To mitigate these risks, I've recommended a set of well-established and practical security measures, including role-based access control, multi-factor authentication, encryption, staff training, and system monitoring. I believe that applying these recommendations would greatly reduce the likelihood and impact of security breaches in hotel environments.

Most importantly, this assessment has reminded me that cybersecurity isn't a one-time fix; it's a continuous process that must evolve with new technologies and emerging threats. For hotel businesses that rely heavily on digital systems and customer trust, proactive security isn't just a technical need; it's a business necessity.

References

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- [2] L. X. a. T. W. Y. Zhou, "DDoS Threats and Countermeasures in the Hospitality Sector," *IEEE Access*, vol. 9, pp. 10091-10104, 2021.
- [3] M. A. N. Z. J. S. Ali, "SQL Injection: A Threat to Online Security," *International Journal of Advanced Computer Science and Applications*, vol. 11, no. 5, pp. 405-410, 2020.
- [4] H. S. A. M. Zulkernine, "TrustGuard: Countermeasure for Phishing Attacks," *Journal of Computer Virology and Hacking Techniques*, vol. 17, no. 2, pp. 117-126, 2021.
- [5] D. Z. a. L. W. Y. Chen, "Ransomware in Hospitality: A Review of Case Studies and Countermeasures," *Journal of*

Cybersecurity, vol. 6, no. 1, pp. 22-34,
2023.

Appendix III: DF Screenshots

Week 8

The screenshot shows a web browser window displaying a Canvas LMS discussion page. The address bar shows the URL https://canvas.acu.edu.au/courses/23977/discussion_topics/189815. The left sidebar contains navigation links: Home, Unit Outline, Modules, Announcements, Discussions (highlighted), Assignments, Assignment Extensions, Quizzes, Grades (with a '2' badge), Reading List, Echo360, Zoom, Study help 24/7 - Studioity, and Help (with a '2' badge). The main content area shows a discussion post by Amir Tamang (AT) dated 11 Jun 9:22. The title is "Introduction (Week 8)". The text discusses the rise of IoT technology in healthcare, its benefits, and the associated cybersecurity risks. It mentions the core idea of the report is to analyze and understand the risks of an IoT-based healthcare system, including wearable sensors, communication devices, cloud platforms, and data analytics tools. It also notes that IoT healthcare systems are particularly vulnerable due to limited resources and lack of built-in security, making them susceptible to threats like data breaches, denial of service (DoS), and man-in-the-middle (MitM) attacks. The post concludes by stating the report uses STRIDE and DREAD security models to categorize risks and assess potential impacts, aiming to help stakeholders understand vulnerabilities and adopt better countermeasures. At the bottom of the discussion area are "Previous" and "Next" buttons. The Windows taskbar at the bottom shows the date as 6/11/2025 and the time as 9:29 AM.

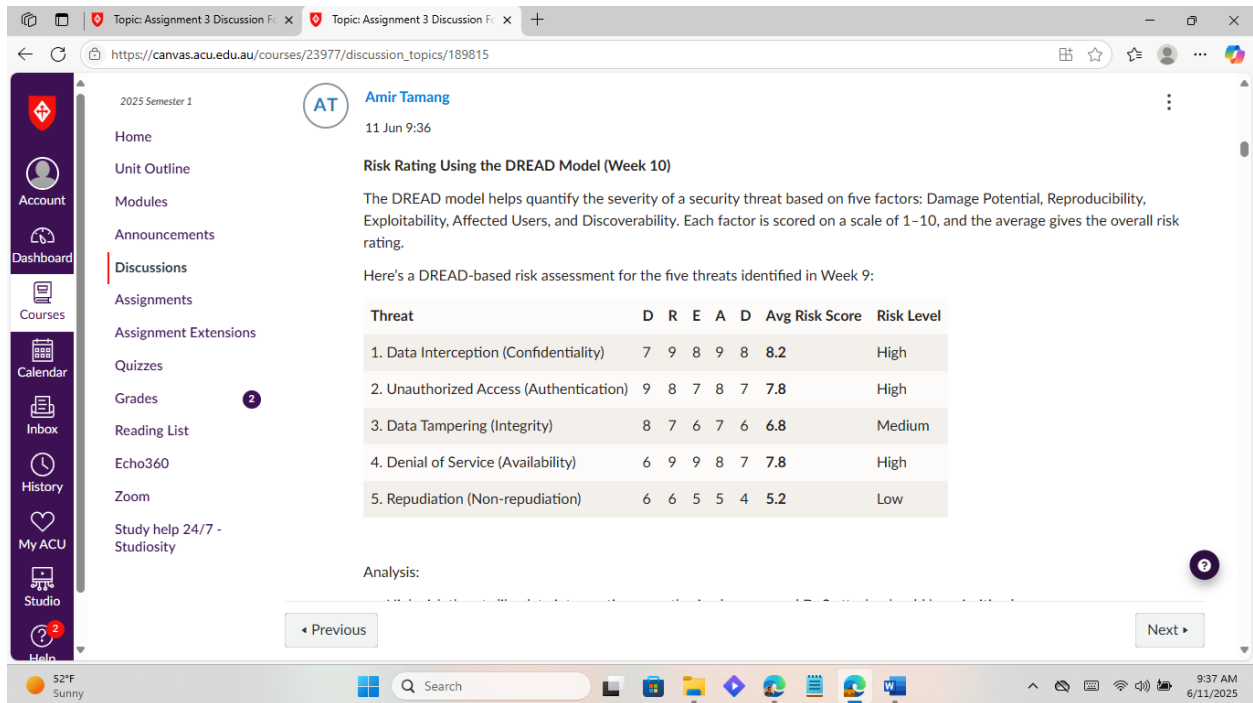
Week 9

The screenshot shows a web browser window displaying a Canvas LMS discussion page. The address bar shows the URL https://canvas.acu.edu.au/courses/23977/discussion_topics/189815. The left sidebar is identical to the Week 8 screenshot, with the "Discussions" link highlighted. The main content area shows a discussion post by Amir Tamang (AT) dated 11 Jun 9:30. The title is "Identify Five Common Security Threats (Week 9)". The text states that in an IoT-based healthcare system, various components are vulnerable to different types of security threats. It lists five common and critical threats that target specific security principles:

- 1. Data Interception and Eavesdropping (Threat to Confidentiality):**
Since IoT healthcare devices transmit patient data over wireless networks (e.g., Bluetooth, Wi-Fi), attackers can exploit unsecured channels to intercept data. Eavesdropping may lead to the exposure of sensitive medical records or identity theft.
Example: In 2020, researchers demonstrated how poorly encrypted Bluetooth communication in some wearable health monitors allowed easy interception of patient data (Sicari et al., 2015).
- 2. Unauthorized Access and Privilege Escalation (Threat to Authentication):**
Weak authentication mechanisms can allow attackers to gain unauthorized access to IoT devices or servers. Once inside, they can modify data, access private records, or even shut down devices.
Example: In 2021, vulnerabilities in medical infusion pumps allowed attackers to alter dosages by bypassing login credentials.
- 3. Data Tampering (Threat to Integrity):**
Attackers can modify patient records or vital sign readings in transit or storage. This can cause life-threatening errors in diagnosis or treatment.
Example: Altered ECG readings in cloud-based health monitoring systems can lead to misdiagnosis (Roman et al., 2013).
- 4. Denial of Service (DoS) Attacks (Threat to Availability):**
DoS attacks target system availability by flooding networks or devices with traffic, preventing legitimate users from accessing the system.

At the bottom of the discussion area are "Previous" and "Next" buttons. The Windows taskbar at the bottom shows the date as 6/11/2025 and the time as 9:31 AM.

Week 10



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AT Amir Tamang

11 Jun 9:36

Risk Rating Using the DREAD Model (Week 10)

The DREAD model helps quantify the severity of a security threat based on five factors: Damage Potential, Reproducibility, Exploitability, Affected Users, and Discoverability. Each factor is scored on a scale of 1–10, and the average gives the overall risk rating.

Here's a DREAD-based risk assessment for the five threats identified in Week 9:

Threat	D	R	E	A	D	Avg Risk Score	Risk Level
1. Data Interception (Confidentiality)	7	9	8	9	8	8.2	High
2. Unauthorized Access (Authentication)	9	8	7	8	7	7.8	High
3. Data Tampering (Integrity)	8	7	6	7	6	6.8	Medium
4. Denial of Service (Availability)	6	9	9	8	7	7.8	High
5. Repudiation (Non-repudiation)	6	6	5	5	4	5.2	Low

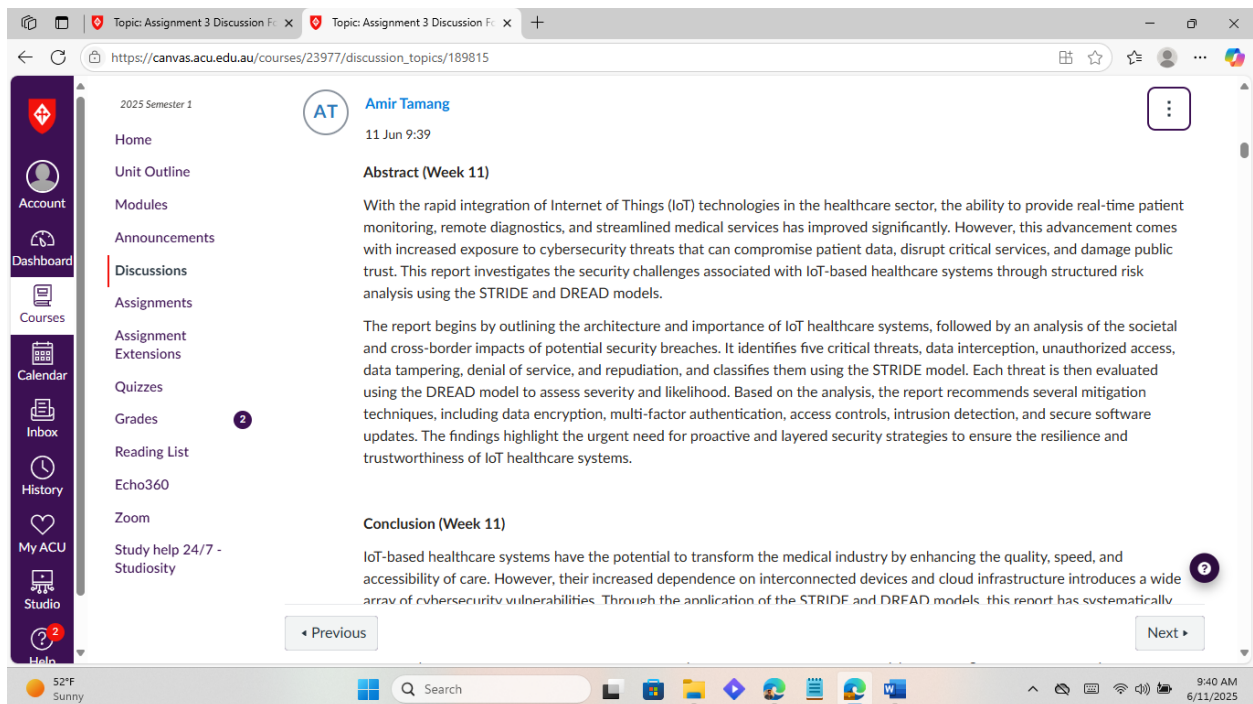
Analysis:

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Week11



2025 Semester 1

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AT Amir Tamang

11 Jun 9:39

Abstract (Week 11)

With the rapid integration of Internet of Things (IoT) technologies in the healthcare sector, the ability to provide real-time patient monitoring, remote diagnostics, and streamlined medical services has improved significantly. However, this advancement comes with increased exposure to cybersecurity threats that can compromise patient data, disrupt critical services, and damage public trust. This report investigates the security challenges associated with IoT-based healthcare systems through structured risk analysis using the STRIDE and DREAD models.

The report begins by outlining the architecture and importance of IoT healthcare systems, followed by an analysis of the societal and cross-border impacts of potential security breaches. It identifies five critical threats, data interception, unauthorized access, data tampering, denial of service, and repudiation, and classifies them using the STRIDE model. Each threat is then evaluated using the DREAD model to assess severity and likelihood. Based on the analysis, the report recommends several mitigation techniques, including data encryption, multi-factor authentication, access controls, intrusion detection, and secure software updates. The findings highlight the urgent need for proactive and layered security strategies to ensure the resilience and trustworthiness of IoT healthcare systems.

Conclusion (Week 11)

IoT-based healthcare systems have the potential to transform the medical industry by enhancing the quality, speed, and accessibility of care. However, their increased dependence on interconnected devices and cloud infrastructure introduces a wide array of cybersecurity vulnerabilities. Through the application of the STRIDE and DREAD models, this report has systematically

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